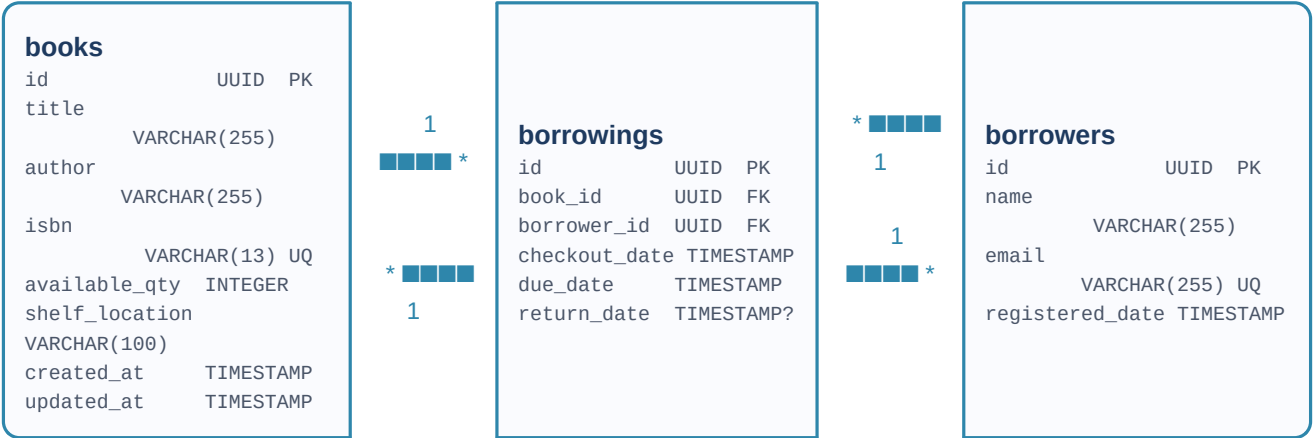


Database Schema

PostgreSQL database with 3 tables managed via TypeORM • Library Management System

Entity Relationship Diagram



Relationships

Relationship	Type	Description
Book → Borrowings	One-to-Many	A book can be borrowed multiple times
Borrower → Borrowings	One-to-Many	A borrower can have multiple active borrowings
Borrowing → Book	Many-to-One	Via book_id foreign key
Borrowing → Borrower	Many-to-One	Via borrower_id foreign key

Table: books

Column	Type	Constraints	Description
id PK	UUID	Auto-generated	Unique book identifier
title	VARCHAR(255)	NOT NULL, trigram indexed	Book title
author	VARCHAR(255)	NOT NULL, trigram indexed	Author name

Column	Type	Constraints	Description
isbn UQ	VARCHAR(13)	NOT NULL, UNIQUE, trigram indexed	ISBN-13 code
available_quantity	INTEGER	NOT NULL, DEFAULT 0	Copies available to lend
shelf_location	VARCHAR(100)	NOT NULL	Physical shelf location
created_at	TIMESTAMP	NOT NULL, auto-set	Record creation time
updated_at	TIMESTAMP	NOT NULL, auto-updated	Last update time

Indexes: `idx_books_title_trgm (GIN)`, `idx_books_author_trgm (GIN)`, `idx_books_isbn_trgm (GIN)`

Table: borrowers

Column	Type	Constraints	Description
id PK	UUID	Auto-generated	Unique borrower identifier
name	VARCHAR(255)	NOT NULL	Borrower full name
email UQ	VARCHAR(255)	NOT NULL, UNIQUE, indexed	Email address
registered_date	TIMESTAMP	NOT NULL, auto-set	Registration date

Indexes: `IDX_borrowers_email (B-tree)`

Table: borrowings

Column	Type	Constraints	Description
id PK	UUID	Auto-generated	Unique borrowing identifier
book_id FK	UUID	FK → books(id), NOT NULL	The borrowed book
borrower_id FK	UUID	FK → borrowers(id), NOT NULL	The borrower
checkout_date	TIMESTAMP	NOT NULL, auto-set	When the book was checked out
due_date	TIMESTAMP	NOT NULL	When the book is due
return_date	TIMESTAMP	NULLABLE	When returned (null = still borrowed)

Indexes: `IDX_borrowings_book_id`, `IDX_borrowings_borrower_id`, `IDX_borrowings_due_date (B-tree)`